PROYECTOS DE INGENIERÍA Y GESTIÓN DEL SOFTWARE

Report identification	
REPORT NUMBER	3
TITLE	WBS and resource allocation
TEAM NUMBER	41
DATE	05/03/2026

Participating members:
Daniel Jesús Morales Sosa
Nataša Đurić
Gligor Gligorov
Luis Martín Pérez

1. Estimation and priority

Number	US	Estimation (story points)	Priority (MoSCoW)	1st Iteration (✔: included)
US1.1	As a user, I want to search for a song to learn by title or artist.	3	M	✅
US1.2	As a user, I want to filter the songs by language.	0,5	C	
US1.3	As a user, I want to see a recommendation based on previous searches.	3	C	
US2.1	As a user, I want to select “singing mode”, where I can be singing along with the lyrics.	3	M	✅
US2.2	As a user, I want my singing to be transcribed into text so that the system can evaluate my accuracy.	13	M	✅
US3.1	As a user, I want to select “fill in the gaps” mode, where i can fill in missing words.	2	M	✅
US3.2	As a user, I want immediate feedback after submitting my answers so that I can learn from my mistakes.	2	S	

US4.1	As a user, I want to see the score obtained on a song after finishing it.	1	S	
US4.2	As a user, I want to see their accumulated score and other obtained achievements on their profile.	1	S	
US5.1	As a user, I want to share the score obtained on a song on Social Networks, as well as my accumulated score and achievements.	3	C	
US6.1	As a user, I want to change the difficulty level. *Note: Available difficulties are Baby, Medium and Hardcore.*	5	C	
US7.1	As a user, I want to save the songs I like to be able to access them faster.	1	S	
US7.2	As a user, I want to log into my account so that I can access my saved songs.	2	S	

T1 (Technical)	The system must integrate a speech-to-text service so that user singing can be transcribed automatically.	5	M	✅
T2 (Technical)	The system must store user events (searches, attempts, scores) in a database.	3	S	
T3 (Technical)	The system must apply difficulty changes during a session by updating gap density/word choice rules.	5	S	
T4 (Technical)	The system must apply different multipliers to calculate the score depending on the difficulty levels.	1	S	
T5 (Technical)	The system must store the following data in the database for each song: audio and captions (in many languages).	3	M	✅

T6 (Technical)	The system must live-stream the song audio to the user.	3	M	✅
T7 (Technical)	The system must permit password recovery for a registered user.	1	C	
T8 (Technical)	The system must have a similarity comparison algorithm between original lyrics and transcribed text so that pronunciation accuracy can be measured.	2	M	✅
T9 (Technical)	The system's frontend must be based on a user-friendly Figma design.	5	M	✅

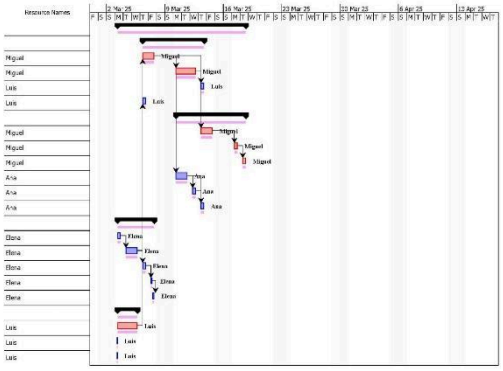
2. Trello board:

https://trello.com/invite/b/69a9ad5d66f026075502fab3/ATTI108c9ca17243ad61d65fe7699e55e96818999573/pigs

3. User story decomposition into tasks and resource allocation (enclose Project Libre file).

		Name	Work	Duration	Start	Finish	Predecessors
1		**Sprint 1**	**184 hours**	**12 days?**	**3/3/25 8:00 AM**	**3/18/25 5:00 PM**	
2		**Search**	**56 hours**	**6 days?**	**3/6/25 8:00 AM**	**3/13/25 5:00 PM**	
3		A user can do a basic simple search by selecting a category	16 hours	2 days?	3/6/25 8:00 AM	3/7/25 5:00 PM	21
4		A user can search for products by entering values in any co...	24 hours	3 days?	3/10/25 8:00 AM	3/12/25 5:00 PM	3
5		A user can order the results of a search by price or rating fr...	8 hours	1 day?	3/12/25 8:00 AM	3/12/25 5:00 PM	3;4
6		A user can see the most frequently acquired products within...	8 hours	1 day?	3/6/25 8:00 AM	3/6/25 5:00 PM	21
7		**Purchase**	**64 hours**	**7 days?**	**3/10/25 8:00 AM**	**3/18/25 5:00 PM**	
8		A user can add products to their shopping cart	16 hours	2 days?	3/13/25 8:00 AM	3/14/25 5:00 PM	4;3
9		A user can remove products from their cart before completin...	8 hours	1 day?	3/17/25 8:00 AM	3/17/25 5:00 PM	8
10		A user can buy the products by entering their billing address...	8 hours	1 day?	3/18/25 8:00 AM	3/18/25 5:00 PM	9
11		A user can open an account that remembers shipping and bil...	16 hours	2 days?	3/10/25 8:00 AM	3/11/25 5:00 PM	3
12		A user can edit the credit card information stored in their acc...	8 hours	1 day?	3/12/25 8:00 AM	3/12/25 5:00 PM	11
13		A user can edit the shipping and billing addresses stored in t...	8 hours	1 day?	3/13/25 8:00 AM	3/13/25 5:00 PM	11;12
14		**Catalog administration**	**40 hours**	**5 days?**	**3/3/25 8:00 AM**	**3/7/25 5:00 PM**	
15		An administrator can add a category into the catalog	8 hours	1 day?	3/3/25 8:00 AM	3/3/25 5:00 PM	
16		An administrator can add a product to the catalog indicating ...	16 hours	2 days?	3/4/25 8:00 AM	3/5/25 5:00 PM	15
17		An administrator can edit the description of a product	8 hours	1 day?	3/6/25 8:00 AM	3/6/25 5:00 PM	16
18		An administrator can delete a product	4 hours	0.5 days?	3/7/25 8:00 AM	3/7/25 1:00 PM	17
19		An administrator can modify the stock of a product	4 hours	0.5 days?	3/7/25 1:00 PM	3/7/25 5:00 PM	18
20		**Technical and constraints**	**24 hours**	**3 days?**	**3/3/25 8:00 AM**	**3/5/25 5:00 PM**	
21		Create Database	24 hours	3 days?	3/3/25 8:00 AM	3/5/25 5:00 PM	
22		Orders made on the website have to end up in the same ord...	0 hours	0 days?	3/3/25 8:00 AM	3/3/25 8:00 AM	
23		The system must support peak usage of up to 50 concurrent...	0 hours	0 days?	3/3/25 8:00 AM	3/3/25 8:00 AM	

Sample PIGS - page1



Instructions:

1. *Using the Estimation and Planning spreadsheet, apply Panning Poker, and assign priorities according to MoSCoW classification.*
2. *Transfer your user stories to a Trello board, select the user stories for the first iteration and copy the link to share the board in this document.*
3. *Transfer the user stories of the first iteration into a ProjectLibre project. Use three levels for your project: iterations, user stories and tasks. You can shorten the titles of user stories, but you must include the Id. Numbers of user stories.*
4. *Set the starting date of Iteration1 to next Monday.*
5. *Make the column Work visible and assign the estimation in it (only for the last level). Although story points do not measure hours, use the conversion rate 1SP=8 hours. The number of hours must be consistent with the estimations.*
6. *Set the dependencies.*
7. *Insert the resources in the RWS and assign the resources to the tasks.*
8. *Check the load of the resources.*
9. *Convert your document into PDF format and name it R3_XX.pdf, where XX is the identifier of your team. Submit it together with the Excel document for estimation and planning, which should be named EP_XX.xlsx, and the ProjectLibre file, which should be named PL_XX.pod, where XX is the identifier of your team.*

4. Use of Artificial Intelligence

Name	Purpose
ChatGPT	Install ProjectLibre on Linux

Instructions:

1. *Using the Estimation and Planning spreadsheet, apply Panning Poker, and assign priorities according to MoSCoW classification.*
2. *Transfer your user stories to a Trello board, select the user stories for the first iteration and copy the link to share the board in this document.*
3. *Transfer the user stories of the first iteration into a ProjectLibre project. Use three levels for your project: iterations, user stories and tasks. You can shorten the titles of user stories, but you must include the Id. Numbers of user stories.*
4. *Set the starting date of Iteration1 to next Monday.*
5. *Make the column Work visible and assign the estimation in it (only for the last level). Although story points do not measure hours, use the conversion rate 1SP=8 hours. The number of hours must be consistent with the estimations.*
6. *Set the dependencies.*
7. *Insert the resources in the RWS and assign the resources to the tasks.*
8. *Check the load of the resources.*
9. *Convert your document into PDF format and name it R3_XX.pdf, where XX is the identifier of your team. Submit it together with the Excel document for estimation and planning, which should be named EP_XX.xlsx, and the ProjectLibre file, which should be named PL_XX.pod, where XX is the identifier of your team.*